

Probability in Plain Language

PHS Launch!

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Goals

Definitions

Operating with probability

Conditional probability

Probability functions

Supplemental slides

Goals:

By the end of this session, you should be familiar with:

- What a “random variable” is, and be able to give some examples
- What a probability represents
- What it means for random variables to be **independent**
- What it means for a random variable to have a specific **distribution**

To make it easier to identify these, I'll highlight them **like this!**

Why is probability useful?

Probability allows us to make statements how likely it is that a given event happens.

For us, probability is a key tool that lets us make **inferences**¹ about relationships between variables - are they coincidental, or something more??

¹see the rest of PHS2000A for more!

Definitions

Sample space

The *sample space* ($\mathcal{S}$) is the set containing all possible outcomes of a trial (sometimes called “experiment”).

Events of interest (e.g., rolling a 6 on a die) can be thought of as a **subset** of this sample space. We denote subsets as $A \subset \mathcal{S}$, where A is one possible outcome of the sample space $\mathcal{S}$.

Sample spaces can be filled with discrete events (e.g. rolling a die) or continuous events (e.g. a person's height)

- *Question:* What does the sample space for flipping a coin contain?

When we consider a sample space, we can create a more math-y **definition of probability** .

Probability is a quantitative (i.e., numerical) measure of how likely $\mathcal{A}$ (some given event in which we are interested) is to occur. We frequently interpret this as the *relative frequency* of $\mathcal{A}$ happening if we were to observe for infinity ², e.g., the proportion of “heads” when flipping a coin infinite times.

Another common definition: the number of “successes”, divided by the number of “trials”.

²Schervish, M. J., & DeGroot, M. H. (2014). Probability and statistics (Vol. 563). London, UK: Pearson Education.

Question:

Why would we quantify probability over an infinite number of attempted events?

Random variable

Mathematically, a **random variable** (RV) *maps the sample space $\mathcal{S}$ a real number $\mathcal{R}$* . An RV can be either **discrete** (integer values) or **continuous** (any real number in a range)

Intuitively, we can think of RVs as being any event or measurement that has some associated probability. The exact value an RV will take isn't known beforehand, *but* we might be able to guess which values might be more or less probable based on previous knowledge of how that RV is **distributed** ³

³more on this later!

Discrete or Continuous?

- Number of astrophage cells on Earth's sun.
- The time it will take astrophage to eat Earth's sun completely (eek!)

Discrete or Continuous?

- Number of astrophage cells on Earth's sun.
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What is the range of each RV?

Game time!

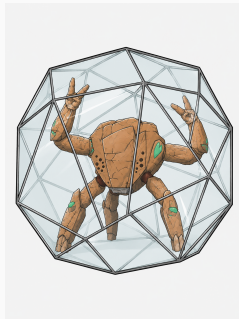
Grace and Rocky don't speak the same language, so they have have very limited shared vocabulary (at first).

Grace says something that Rocky doesn't quite understand, so Rocky decides to just say some random word/phrase using his limited vocabulary. Let's say Rocky knows the following words and is equally likely to say any of them:

bad	amaze!	question	Eridian	observe
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What is the probability that Rocky says "amaze!"?

Bonus question: What is our sample space here? What is our event? What is our random variable?



Operating with probability

AND and OR

When we conceive of probabilities as taking place in a sample space, it's easy to use some basic set operators to think about how we can combine the probabilities of multiple events.

AND

When we are interested in the probability of the co-occurrence of two (or more!) events of interest, we look at their **intersection**.

The intersection of A and B is the subset of the sample space where both A **and** B occur.

We denote intersections with the symbol $\cap$.

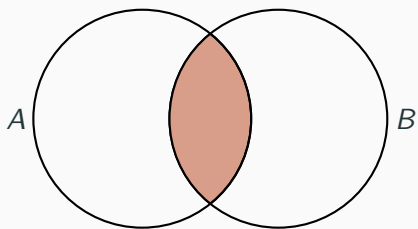
OR

When we are interested in whether at least one of a subset of events of interest occur, we look at their **union**.

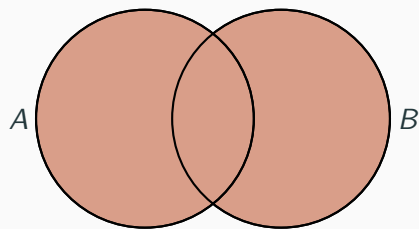
The union of A and B is the subset of the sample space where A **or** B **or** both occur.

We denote unions with the symbol $\cup$.

AND and OR: Venn diagrams



$A \cap B$



$A \cup B$

Finding $P(A \cup B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Why? We look at the probability of each event happening on their own, then subtract the probability of both events happening to avoid double-counting those scenarios.

Finding $P(A \cap B)$

$$\begin{aligned}P(A \cap B) &= P(B) \times P(A|B) \\ &= P(A) \times P(B|A)\end{aligned}$$

Why? We look at the probability of the first event happening, **then** check the probability of the second event happening *given that* the first event happened.

For example, the probability of drawing a heart from a deck of cards, then drawing another heart, is $P(\text{heart} \cap \text{heart}) = \frac{12}{52} \times \frac{11}{51} = 0.05$

Game time! AND and OR

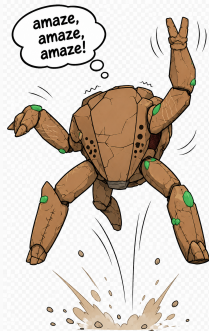
Rocky is getting better at understanding Grace, and has come up with some generally affirmative phrases he can use to encourage Grace when times get tough:^a

amaze amaze amaze!	good good yes	amaze happy happy	amaze good good	yes yes yes
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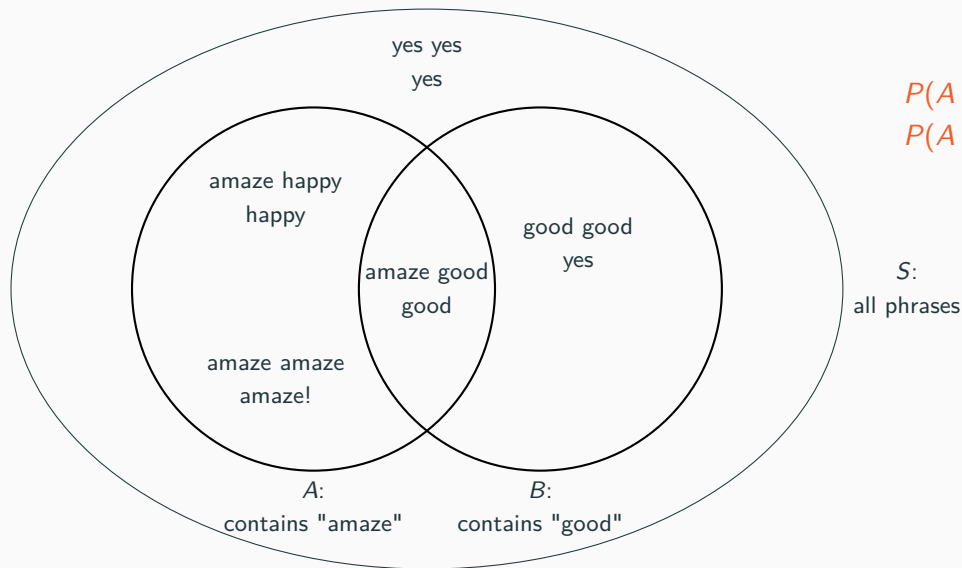
If Rocky chooses a response at random, what is the probability that it contains "amaze" **and** "good"?

If Rocky chooses a response at random, what is the probability that it contains either "amaze" **or** "good" (or both)?

^amay we all have a Rocky in our tough times...



Game time! AND and OR



Disjoint Events (Mutually Exclusive)

Two events A and B are **disjoint**, or **mutually exclusive**, if both A and B happening together is impossible. For example, a flipped coin cannot land on both heads and tails on the same flip.

When two events are disjoint, $P(A \cap B) = 0$.

What are some examples of this from your research/public health?

Independent Events

Two events A and B are **independent** if the probability of A occurring does not impact the probability of B occurring (and vice versa). For example, a coin landing on 'heads' has no impact on how it will land in a later flip.

This is denoted as $A \perp B$.

If $A \perp B$, then $P(A|B) = P(A)$ and $P(B|A) = P(B)$. Also, $Pr(A \cap B) = P(A) \times P(B)$

What are some examples of this from your research/public health?

Question: are mutually exclusive events independent?

Independent events: A disclaimer

The rule $P(A) \times P(B) = Pr(A \cap B)$ is sometimes used to define independent events, but also as a property that is invoked after claiming two events are independent. This can quickly get us into a circular logic issue.

Independence is a property we need to assume for a lot of the models and methods we use in quantitative research to hold, but it's not something we can always check with this rule!

Conditional probability

Conditional vs Marginal

A few slides back, we sneakily introduced the concept of **conditional** probability. Now we want to make that piece explicit. A probability is conditional when we are concerned with the probability of an event **among** some group with a particular characteristic, or **among** a group who experiences a second event of interest. In notation, a conditional probability is denoted $P(A|B)$.

Conditional probabilities are common in the quantitative methods learned in this course. Sometimes conditional probabilities are necessary to make the methods “work” and remove bias; sometimes conditional probabilities are of scientific interest in their own right.

Game Time! Conditional Probabilities.

Rocky has a pretty large vocabulary now, but Grace still relies on using Rocky's computer translator, which limits Rocky's ability to use tone. One way he compensates is by saying “question” whenever he needs Grace to answer something.

Because a lot of Rocky's question are about humans, Grace figures that any time Rocky starts a statement with “you”, there's a good chance he'll ask a question. . . but how likely is that, actually?

Open the `prob_with_rocky.R` and find out!

Hint 1: What is A ? What is B ?

Hint 2: We can rearrange our equation from slide 16 to find $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Conditional probabilities: Public health examples

Unconditional

- probability of being hospitalized with COVID-19
- probability of death after a heart attack
- probability of developing childhood asthma

Conditional

- probability of being hospitalized with COVID-19 **among** those who are vaccinated
- probability of death after a heart attack **among** those over age 70
- probability of developing childhood asthma **among** those living in historically redlined neighborhoods

In you own words: what is the difference between an intersection and a conditional probability?

Or, what is the difference between $P(A \cap B)$ and $P(A|B)$?



Flipping the conditioning statement

Do you think that $P(A|B) = P(B|A)$?

Flipping the conditioning statement

No! Generally speaking, $P(A|B) \neq P(B|A)$.

This stinks, because sometimes we know $P(A|B)$, and are interested in finding $P(B|A)$. What can we do here?

But wait! Statistics has a solution

Bayes' Theorem

Bayes' Theorem allows us to go from $P(A|B)$ to $P(B|A)$. Let's start with two things we know and do a small derivation:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A|B) \cdot P(B) = P(A \cap B)$$

$$\text{Thus, } P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)} !$$

(A more general form of this theorem is shown in the supplemental slides, for anyone interested)

Probability functions

Probability Mass/Density Function

Definition: Let X be a random variable. A probability function is any function $f(x)$ such that for every possible value that X can take, $f(x) = P(X = x)$. Essentially, it's a function that tells us how likely certain values are.

The range and relative probabilities of values that an RV can take on are referred to as its **distribution**. There are many common distributions we will use in this class, each with an associated probability function.

A Probability **Mass** Function (PMF) is for discrete random variables, and a Probability **Density** Function (PDF) is for continuous random variables.

Quick reminder: What is a discrete random variable?

Probability Mass Function: Bernoulli

The simplest distribution we work with in this class is the **Bernoulli** distribution, which characterizes binary probabilities (i.e., either a “success” or a “failure”).

When they first meet, Rocky throws Grace 2 canisters with a message inside, and Grace manages to catch 1 of them. Let's say this means there's a $\text{\textbf{10\%}}$ chance that Grace catches any given message, and notate it as:

$$X = \begin{cases} 0 & \text{Grace catches it} \\ 1 & \text{Grace misses it} \end{cases}$$

We say that X is **distributed as Bernoulli with $p = 0.1$** , or, $X \sim \text{Bern}(p = 0.1)$.

We could calculate the probability of either outcome using the function:

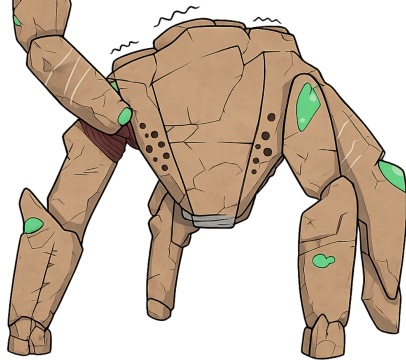
$$P(X = x) = p^x(1 - p)^{1-x}$$

Probability Density Function

PDFs, used for continuous RVs, differ from PMFs in one key way: instead of quantifying *probability*, they describe the *relative likeliness* of a given outcome. This is because the probability of a continuous variable taking on a specific value *exactly* is very very small (actually, 0!) so PDFs describe how probable certain values are relative to other values.

The other very common distribution we will work with is the Normal distribution - a very handy distribution that is rather complex mathematically, but describes a shocking number of natural phenomena. It's particularly handy for inference and calculating confidence intervals.

((YOU ARE



AMAZE AMAZE
AMAZE!

Thank you!

Thank you to past PHS2000 TFS, in particular Logan Beyer & Nicky Rahim, for creating past versions of this slidedeck.

Supplemental slides

Key distributions

Bernoulli and binomial distributions

Both describe trials of events with probability p of success.

Bernoulli distribution

- Describes a single trial of an event with probability p .
- The probability distribution defines the probability that the single trial is a "success".

Binomial distribution

- Describes n trials of an event with probability p , where n is an integer.
- The probability distribution defines the probability that k out of the n trials are a "success".

Bernoulli and binomial distributions

Bernoulli distribution

- Parameterized with p alone.
- The PMF takes the form

$$P(X = k) = p^k(1 - p)^{1-k}$$

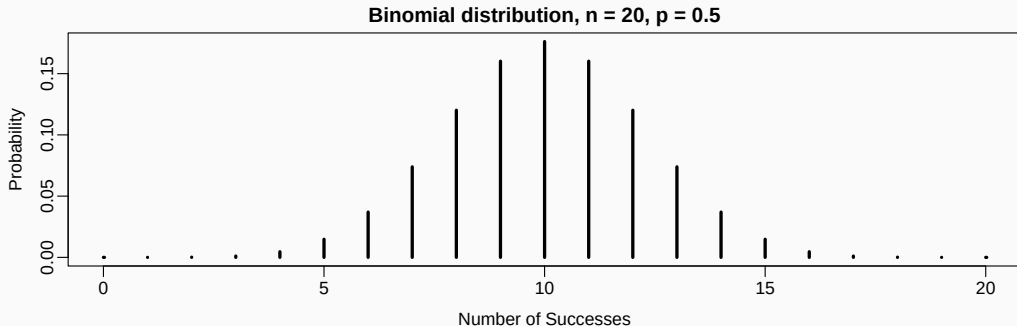
where $k \in \{0, 1\}$

Binomial distribution

- Parameterized with p and n .
- The PMF takes the form

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

where $k \in \{0, 1, 2, \dots, n\}$.



Poisson distribution

The Poisson distribution is used to describe the distribution of **count** events.

Examples may include - Traffic accidents at a particular intersection - CVD events in a population - Times my cat knocks a pen off the table

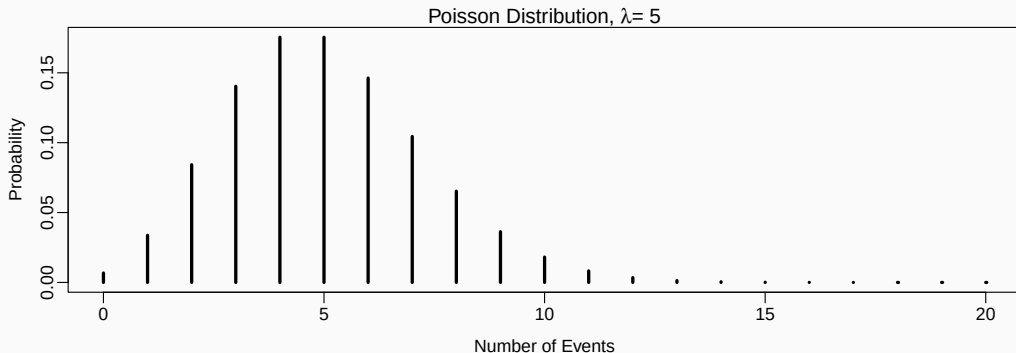
Processes that give rise to events that follow a Poisson distribution are often called **Poisson processes**.

When we consider counts generated by a Poisson process occurring **over time**, Poisson distributions can help us to understand and model event **rates**.

Poisson distribution

The Poisson distribution can be fully described by the parameter λ , which is both the expected count (mean) and the variance of the distribution.

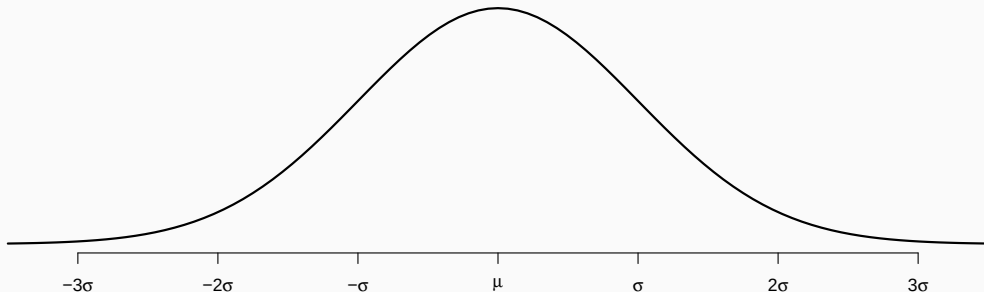
The PMF takes the form $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$ where $k \in \{0, 1, 2, \dots, n\}$



Normal distribution

Describes continuous data with values spread evenly to both sides of the mean

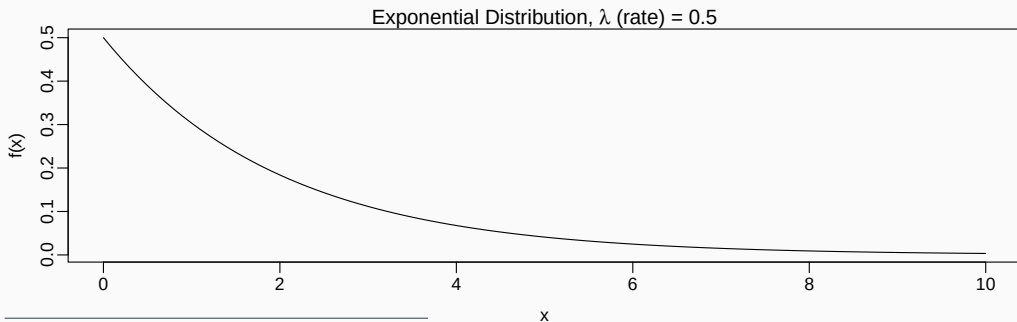
- Can be completely defined with mean μ and variance σ^2 .
- Possible values range from $-\infty$ to ∞ (although there is very little density in the tails!)
- The PDF takes the form $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$



Exponential distribution

The exponential distribution describes the time between events, where the events are generated by a Poisson process. The distribution can be parameterized by a single parameter λ , which here denotes the rate (hazard) at which events occur.⁴

The PDF takes the form $f_X(x) = \lambda e^{-\lambda x}$ where $x \in (0, \infty)$



⁴See the survival lectures later this fall for more on the hazard!

Useful probability rules

Bayes' Theorem

If $A_1, A_2, \dots, A_n$ are mutually exclusive and exhaustive subsets of a sample space (meaning at least one of $A_1, \dots, A_n$ must always occur), then for each value $i \in \{1, \dots, n\}$,

$$P(A_i|B) = \frac{P(B|A_i) \times P(A_i)}{\sum_{j=1}^n P(B|A_j) \times P(A_j)}$$

Note that we call $A_1, A_2, \dots, A_n$ a **partition of the sample space**, and notate it as $\bigcup_{i=1}^n A_i = S$.

Law of total probability

The law of total probability (or total expectation) allows us to “condition out” or “marginalize over” a variable that we do not want to condition on.

$$P(A) = \sum_x P(A|X = x) \times P(X = x)$$

This also holds for conditional probabilities.

$$P(A|B) = \sum_x P(A|B, X = x) \times P(X = x|B)$$

In the notation above, $\sum_x$ is commonly used shorthand for “sum over all possible values of the random variable $X = x$ ”. If X is a standard six-sided dice, $\sum_x$ would mean summing over the values $X = 1, X = 2, \dots, X = 6$, for example.